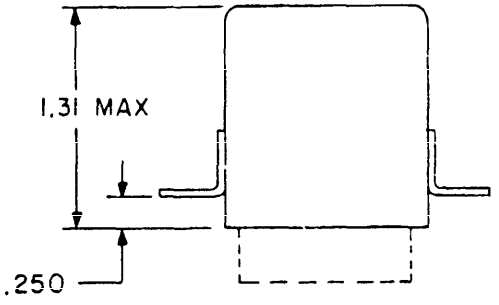
MILITARY SPECIFICATION SHEET

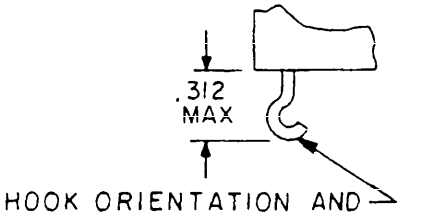
RELAY, ELECTRICAL, HERMETICALLY SEALED, DPDT, 10 AMPERES

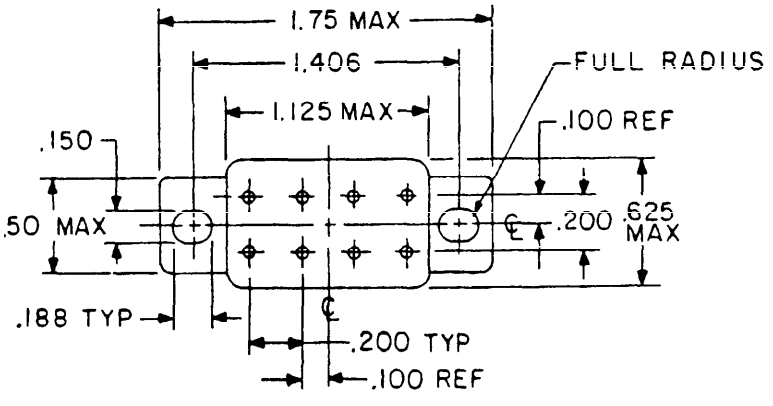
© INACTIVE FOR NEW DESIGN AFTER 18 February 1986
USE MIL-R-5757/23-001

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

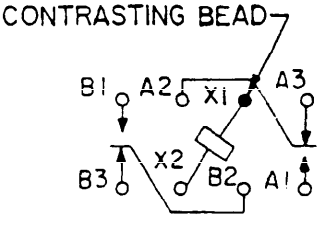
The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-5757.







LUG



© CIRCUIT DIAGRAM
COIL DEENERGIZED
(TERMINAL VIEW)

Inches	mm	Inches	mm
.100	2.54	.50	12.7
.150	3.81	.625	15.88
.188	4.78	1.125	28.58
.200	5.08	1.31	33.4
.250	6.35	1.406	35.71
.312	7.95	1.75	44.5

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified tolerances are ±.010 (0.25 mm) for three place decimals.
4. Terminal numbers need not appear on the relay header provided there is affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
© 5. Circuit diagram shown on part is the terminal view.

FIGURE 1. Dimensions and configuration.

© denotes changes

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REQUIREMENTS:

© Contact data: (CAUTION: High-level characteristics relays are not designed or intended for use in low level applications).

Configuration: DPDT.

Arrangement: 2 form C.

Load ratings: (Relay case grounded)

Resistive: 10 amperes, 28 V dc.

Minimum current: Not applicable.

Contact voltage drop:

Rated life:

Before: 0.1 V dc maximum.

During: 10 percent of open circuit voltage maximum.

After: 0.2 V dc maximum.

Contact bounce: 2 milliseconds (ms) maximum.

Overload:

Resistive: 2 times rated current.

Coil data:

Duty rating: Continuous.

Maximum voltage: 32 V dc.

Nominal voltage: 26.5 V dc.

Pickup voltage: 18 V dc maximum over the temperature range.

Hold voltage: 14 V dc over the temperature range.

Dropout voltage: 1.4 V dc over the temperature range.

Coil resistance: 300 ohms ±10 percent.

Operate time: 15 ms maximum over temperature range.

Release time: 10 ms maximum over temperature range.

Electrical data:

Insulation resistance: 1,000 megohms minimum except the resistance between coil and case at high temperature shall be 500 megohms or greater.

Dielectric withstanding voltage:

	Sea level V rms (60 Hz)
Between case, frame, or enclosure and all contacts:	1,000
Between case, frame, or enclosure and coil:	500
Between all contacts and coil:	1,000
Between open contacts in the energized and deenergized positions:	500
Between contact poles:	1,000

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Environmental data:

Temperature range: Symbol B (-65°C to +125°C).

Internal moisture: Applicable.

Vibration: Symbol 2 (10 g's, 10 to 500 Hz). A resonant frequency shall not exist below 500 Hz.

Shock: Symbol 3 (100 g's).

Physical:

Terminal strength: 3 ±0.3 pound pull.

Terminal solderability: Applicable.

Sealed by welding: Applicable.

Dimensions and configuration: See figure 1.

Terminations: Solder lug.

Weight: 2.0 ounces, maximum.

Life test requirements:

High level: 100,000 cycles at rated load.

© Part number: M5757/28-001

Quality assurance:

© Periodic inspection: Groups B and C inspections are not applicable.

© Group A.

1. Subgroup 2 - 100 percent
2. Dielectric withstanding voltage:
   (a) Tests to be conducted at sea level rating only
   (b) Duration of application - 5-10 seconds at a 10 percent increase in the dielectric strength voltage.
3. Subgroup 3 - 100 percent

Qualification inspection. Single submission only is applicable. The qualification sample size shall be 17 units including one open unit. The sample units shall be subjected to all specified tests in the qualification inspection table of MIL-R-5757.

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Qualification inspection (reduced testing) and sample size: See table I. If the relays produced for MIL-R-5757/28 are similar in construction and design to the 26.5 V dc rated coil voltage of the relays produced for MIL-R-5757/23 then reduced testing for qualification of MIL-R-5757/28 relays may be performed concurrent with or subsequent to successful qualification of MIL-R-5757/23.

TABLE I. Qualification inspection (reduced testing).

Examination or test
6 units shall be subjected to Group I of qualification inspection table and life testing.
1 unsealed sample unit for internal inspection.

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5945-N607)

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